

Assignment 1- CP363

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Group # 26

Equipment Test Center

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Introduction

This project focuses on designing a database application for managing laboratory equipment testing and client bookings. The system is intended to streamline the operations of a testing facility by organizing information about clients, technicians, equipment, laboratories, and test activities. To achieve this, ten entities with appropriate attributes and constraints have been defined, along with weak entities to capture dependent data such as invoices and maintenance records. The model also establishes meaningful relationships between entities, ensuring that key interactions, such as clients making bookings, technicians supervising test sessions, and equipment undergoing maintenance, are clearly represented. Business rules have been specified to maintain consistency and accuracy, while participation and cardinality constraints define how entities interact. An ER diagram visually illustrates the structure, capturing the entities, relationships, constraints, and weak entities, forming a comprehensive foundation for database implementation.

Entities with Attributes

1. Equipment

- Equipment_ID (PRIMARY KEY, NOT NULL, UNIQUE)
- Name (VARCHAR, NOT NULL)
- Model_Number (VARCHAR, NOT NULL)
- Manufacturer (VARCHAR, NOT NULL)
- Status (VARCHAR, NOT NULL)

2. Client

- Client_ID (PRIMARY KEY, NOT NULL, UNIQUE)
- First_Name (VARCHAR, NOT NULL)
- Last_Name (VARCHAR, NOT NULL)
- Contact_Email (VARCHAR, NOT NULL, UNIQUE)
- Phone_Number (VARCHAR, NOT NULL, UNIQUE)
- Organization (VARCHAR, NOT NULL)

3. Technician

- Technician_ID (PRIMARY KEY, NOT NULL, UNIQUE)
- First_Name (VARCHAR, NOT NULL)
- Last_Name (VARCHAR, NOT NULL)
- Certification (VARCHAR, NOT NULL)
- Contact_Email (VARCHAR, NOT NULL, UNIQUE)
- Phone_Number (VARCHAR, NOT NULL, UNIQUE)

4. Test

- Test_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Test_Name (VARCHAR, NOT NULL)
 - Description (VARCHAR, NOT NULL)
5. **Test_Session** (when a test is actually conducted)
- Session_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Test_Date (DATE, NOT NULL)
 - Result (VARCHAR, NOT NULL)
 - Duration (TIME, NOT NULL)
6. **Booking**
- Booking_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Booking_Date (DATE, NOT NULL)
 - Client_ID (FOREIGN KEY, NOT NULL)
 - Lab_ID (FOREIGN KEY, NOT NULL)
 - Start_Time (TIME, NOT NULL)
 - End_Time (TIME, NOT NULL)
 - Status (VARCHAR, NOT NULL)
7. **Lab**
- Lab_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Lab_Name (VARCHAR, NOT NULL)
 - Location (VARCHAR, NOT NULL)
 - Capacity (INTEGER, NOT NULL)
8. **Report**
- Report_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Report_Date (DATE, NOT NULL)
 - Findings (VARCHAR, NOT NULL)
 - Approved_by (INTEGER, NOT NULL, FOREIGN KEY = Technician_ID)
9. **Invoice** (Weak Entity, depends on the booking)
- Invoice_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Invoice_Date (DATE, NOT NULL)
 - Booking_ID (FOREIGN KEY, NOT NULL)
 - Amount (FLOAT, NOT NULL)
 - Payment_Status (BOOLEAN, NOT NULL)
10. **Maintenance Record** (Weak Entity, depends on Equipment)
- Record_ID (PRIMARY KEY, NOT NULL, UNIQUE)
 - Equipment_ID (FOREIGN KEY, NOT NULL)
 - Maintenance_Date (DATE, NOT NULL)
 - Technician_ID (FOREIGN KEY, NOT NULL)

- Notes (VARCHAR, NOT NULL)

Relationships

1. Client and Booking

- Relationship: A client can make many bookings; each booking belongs to one client.
- Cardinality: (1:N)
- Participation:
 - ➔ Customer; Partial - All clients may not have a booking
 - ➔ Booking; Full - All bookings are by a client

2. Booking and Equipment

- Relationship: A booking may reserve multiple equipment items; equipment can appear in multiple bookings.
- Cardinality: (M:N)
- Participation:
 - ➔ Booking; Full - All bookings are for some equipment(s)
 - ➔ Equipment; Partial - All equipments may have a booking

3. Booking and Lab

- Relationship: Each booking is assigned to exactly one lab; a lab can host many bookings.
- Cardinality: (1:N)
- Participation:
 - ➔ Booking; Full - All bookings may have a Lab
 - ➔ Lab; Partial - All Labs may not have a Booking

4. Technician and Test_Session

- Relationship: A test session must be supervised by a technician; a technician can supervise many sessions.
- Cardinality: (1:N)
- Participation:
 - ➔ Technician; Partial - All Technicians may not supervise the Test_Session
 - ➔ Test_Session; Full - All Test_Sessions will be supervised by a Technician

5. Test and Test_Session

- Relationship: Each test session corresponds to one test type; a test can have many sessions.
- Cardinality: (1:N)
- Participation:
 - ➔ Test; Partial - All the Test types cannot be in one Test_Session
 - ➔ Test_Session; Full - All Test_Sessions will have a Test

6. Equipment and Test_Session

- Relationship: A test session may involve one or more equipment items; equipment can be tested in multiple sessions.
- Cardinality: (M:N)
- Participation:
 - Equipment; Partial - All Equipments may not go through a test session
 - Test_Session; Full - All Test_Sessions have an Equipment

7. **Test_Session and Report**

- Relationship: Each session produces one report; each report belongs to exactly one test session.
- Cardinality: (1:1)
- Participation:
 - Test_Session; Full - All Test_Sessions will produce a report
 - Report ; Full - All reports are produced in a Test_Session

8. **Client and Invoice**

- Relationship: A client may have many invoices; each invoice belongs to one client.
- Cardinality: (1:N)
- Participation:
 - Client; Partial - All clients may not have an invoice
 - Invoice; Full - All the Invoices belongs to a Client

9. **Booking and Invoice**

- Relationship: Each booking generates one invoice; each invoice corresponds to one booking.
- Cardinality: (1:1)
- Participation:
 - Booking; Full - All booking generates an Invoice
 - Invoice; Full - All Invoices corresponds to a booking

10. **Equipment and Maintenance_Record**

- Relationship: An equipment may have multiple maintenance records; No maintenance record can exist without equipment.
- Cardinality: (1:N)
- Participation:
 - Equipment; Partial - All the Equipment may not have a Maintenance_Record
 - Maintenance_Record; Full - All the Maintenance_Records belong to some Equipment.

Business Rules

1. Clients must register before making equipment test bookings.
2. Each booking must specify at least one piece of equipment and is assigned to exactly one lab.

3. Every booking generates exactly one invoice.
4. A technician must supervise every test session.
5. Each test session involves one predefined test and may use one or more equipment items.
6. Each test session produces exactly one report.
7. Equipment must undergo regular maintenance, and each maintenance record is tied to one equipment (weak entity).
8. A lab can host multiple bookings, but equipment can only be used in one booking at the same time.
9. A client can have multiple invoices, but each invoice belongs to exactly one booking.

ER/ EER Diagram

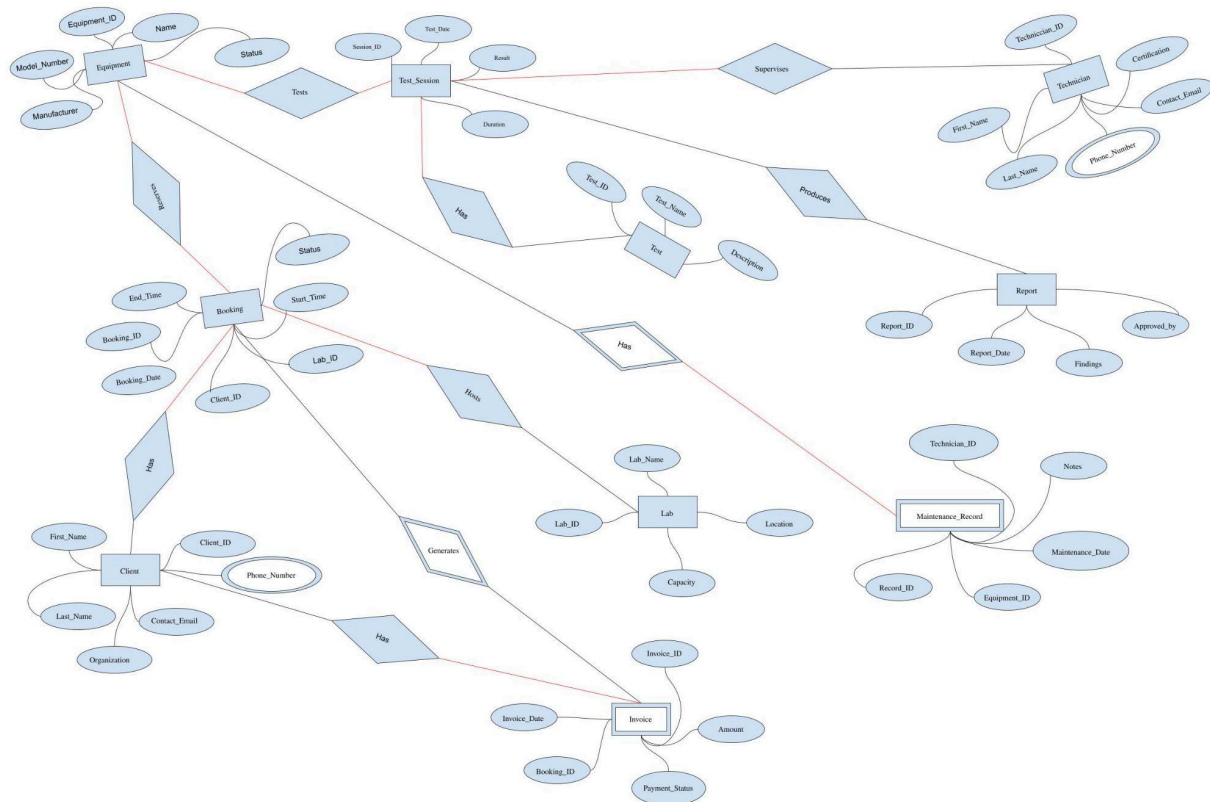


Diagram Key to Read

- Red - Many Relations
- Black - Single/ One Relation